SYDNEY GRAMMAR SCHOOL

2021
TRIAL EXAMINATION

BIOLOGY

Form VI

STRUCTURE OF PAPER

SECTION I - 20 marks (pages 3-13)

- Attempt Questions 1- 20
- Allow about 35 minutes for this section

SECTION II – 80 marks (pages 15-39)

- Attempt Questions 21- 30
- Allow about 2 hours and 25 minutes for this section

EXAMINATION

DATE:	Tuesday 24th August 8.40 AM
DURATION:	3 hours + 5 minutes reading
MARKS:	100

CHECKLIST

Each boy should have the following:

☐ 1 Examination Paper

☐ 1 Multiple-Choice Answer Sheet

GENERAL EXAM INSTRUCTIONS

- 5 minutes reading time and 3 hours working time.
- Write using black pen and draw diagrams using pencil.
- For questions in Section II, show all relevant working for questions involving calculations.
- NESA approved calculators may be used.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- WRITE YOUR **CANDIDATE NUMBER** IN THE SPACE PROVIDED AT THE TOP OF EACH SECTION AS INDICATED IN SECTION II.

CLASS NUMBER	1	2	3	4
Class	6BL201	6BL202	6BL203	6BL204
Master Initials	ZI	HCKM	DBD	ACR

Authors: ACR, DBD, HCKM, ZI   MiC: ACR

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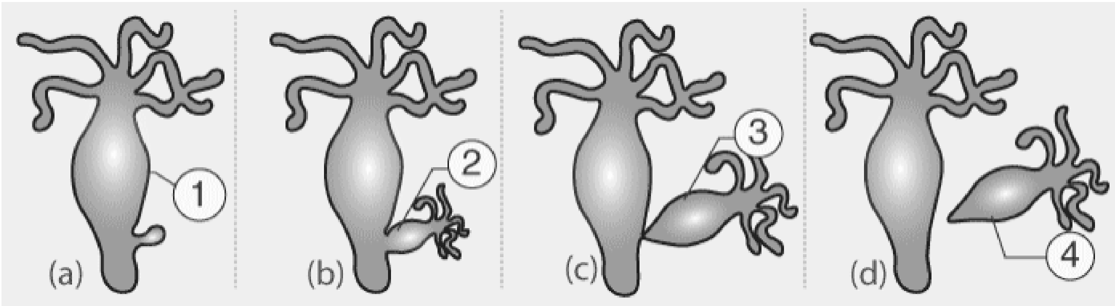
SECTION I: MULTIPLE CHOICE (20 marks)

Attempt ALL Questions
Use the Multiple-Choice Answer Sheet.

1 The following diagram shows the reproduction process in Hydra (an aquatic multicellular organism).



What is the correct name given to this type of reproduction?

(A) Fusion

(B) Fission

(C) Mitosis

(D) Budding

2 The following shows the DNA template strand for a coding section of DNA.

A T C G G C T A T

In the double stranded molecule of DNA formed from this template, how many nucleotides would be present?

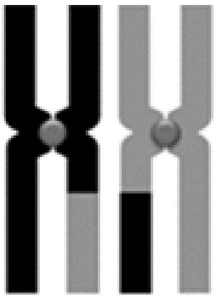
(A) 3

(B) 9

(C) 10

(D) 18

3 The following image shows a process that may occur during a stage of cell division.



Which type of cell division would this process be seen in?

(A) Mitosis

(B) Meiosis

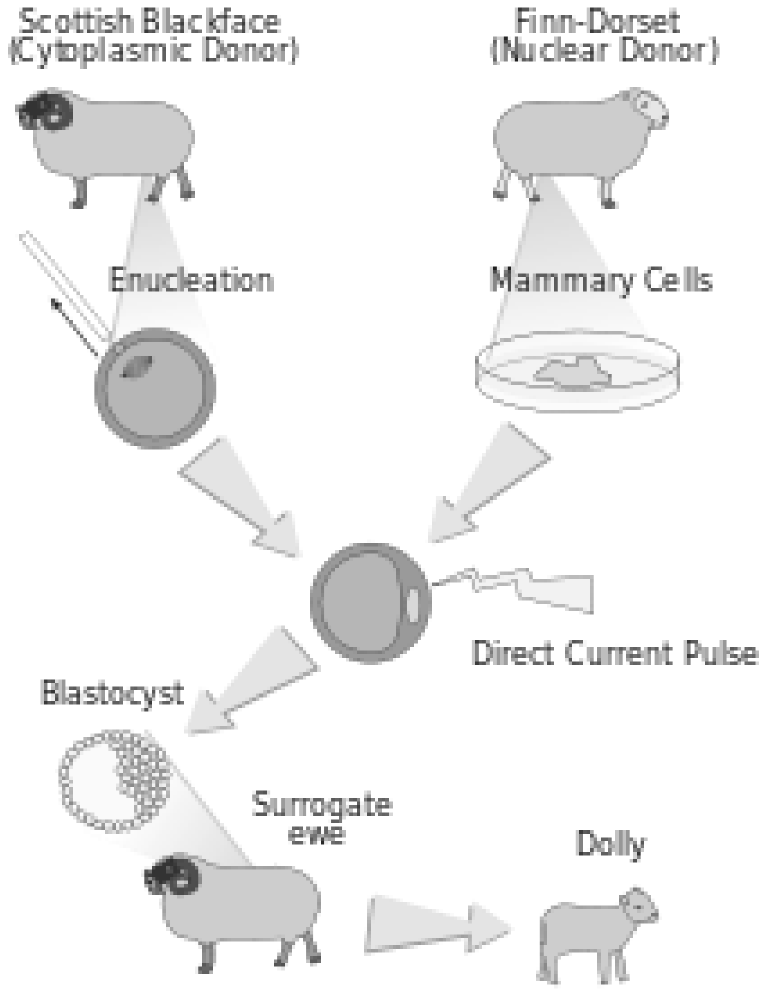
(C) Cloning

(D) Fertilisation

4 Which row indicates the correct role for each of the hormones shown?

	LH	**FSH**	**Progesterone**	**Oestrogen**
(A)	Causes the mature egg to be released from the ovary	Prepares the uterus to receive a fertillised ovum	Stops FSH being produced	Maintains the lining of the uterus during the middle of the menstrual cycle
(B)	Maintains the lining of the uterus during the middle of the menstrual cycle	Causes an egg to mature in the ovary	Stimulates the pituitary gland to release LH	Causes the mature egg to be released from the ovary
(C)	Causes the mature egg to be released from the ovary	Causes an egg to mature in the ovary	Prepares the uterus to receive a fertillised ovum	Stimulates the pituitary gland to release LH
(D)	Stimulates the ovaries to release oestrogen	Prepares the uterus to receive a fertillised ovum	Stops FSH being produced	Maintains the lining of the uterus during the middle of the menstrual cycle

5 The following diagram shows the process of cloning in sheep.



Which of the following best describes the likely appearance of the lamb (Dolly) produced by this process?

(A) Dolly will look like the Finn-Dorset sheep.

(B) Dolly will look like the Scottish Blackface sheep.

(C) Dolly will look like neither the Scottish Blackface nor the Finn-Dorset sheep.

(D) Dolly will look like a mixture of the Scottish Blackface and Finn-Dorset sheep.

6 Giardiasis is an intestinal infection caused by the Giardia parasite, a protozoan which can be found in animal and human faeces. These parasites also thrive in contaminated food, water, and soil, and can survive outside a host for long periods of time.

The best strategy for controlling the incidence of giardiasis is the use of which of the following?

(A) Antibiotics.

(B) Quarantine.

(C) Social distancing by two arm lengths.

(D) Washing your hands with soap for 20 seconds.

7 A pathogen is found to have circular chromosomes and a cell wall.

How would it be classified?

(A) virus

(B) fungus

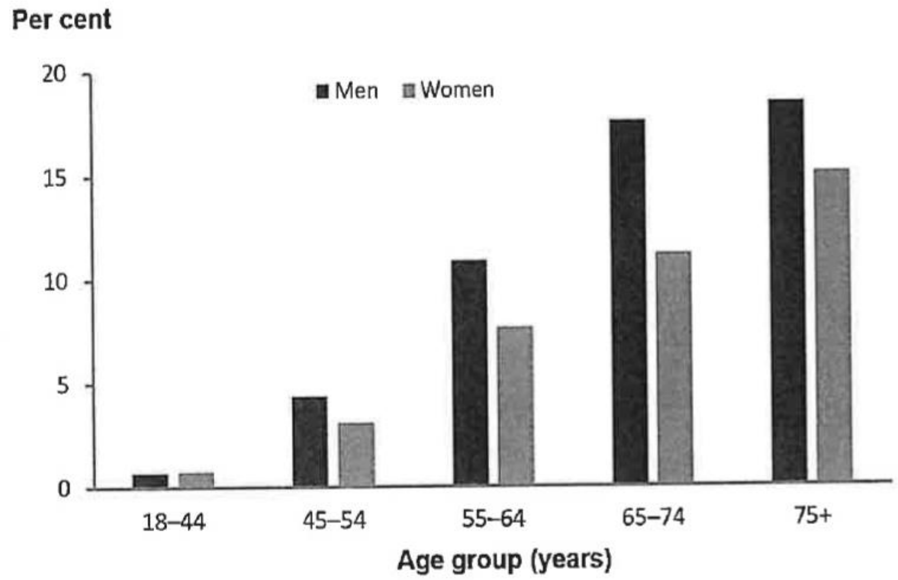
(C) bacterium

(D) protozoan

Refer to the following information to answer **questions 8 and 9**.

An estimated 1.5 million (out of 24.6 million) Australians had Type 2 Diabetes in 2017-2018.

Figure 1: Prevalence of self-reported type 2 diabetes, among persons aged 18 and over, by age and sex, 2017–18



Sources: Australian Institute of Health & Welfare, Australian Bureau of Statistics & Diabetes NSW

8 According to the data, what was the approximate difference in prevalence of self-reported type 2 diabetes between men 55-64 years and men 75+ years?

(A) 7%

(B) 10.5%

(C) 14%

(D) 17.5%

9 What was the prevalence of Type 2 Diabetes in Australia in 2017-2018?

(A) 6.1 per 100,000 persons

(B) 610 per 100,000 persons

(C) 6098 per 100,000 persons

(D) 60976 per 100,000 persons

10 A new treatment for a disease is developed that prevents death but does not produce recovery from the disease. Which of the following will occur?

(A) Incidence will increase.

(B) Incidence will decrease.

(C) Prevalence will increase.

(D) Prevalence will decrease.

11 National Health Surveys are conducted every 3-4 years to gather data on the health of individuals from selected households across Australia. This data is used to identify rates of disease and gather information on health-related factors, for instance smoking. Approximately 25,000 households are requested to complete a detailed questionnaire with trained interviewers.

What would most improve the reliability of this method?

(A) Conducting the survey every 5 years.

(B) Increasing the number of households surveyed.

(C) Using another country to compare the data against.

(D) Allowing households to self-report rather than being interviewed.

12 Which of the following would produce antibodies?

(A) T cell

(B) Bacteria

(C) Killer cell

(D) Plasma cell

13 Xerophytes are plants that are adapted to dry and arid conditions. Xerophytes have physiological and structural adaptations to maximise water conservation. Which of the following adaptations would help to conserve and therefore maintain water balance in a xerophyte?

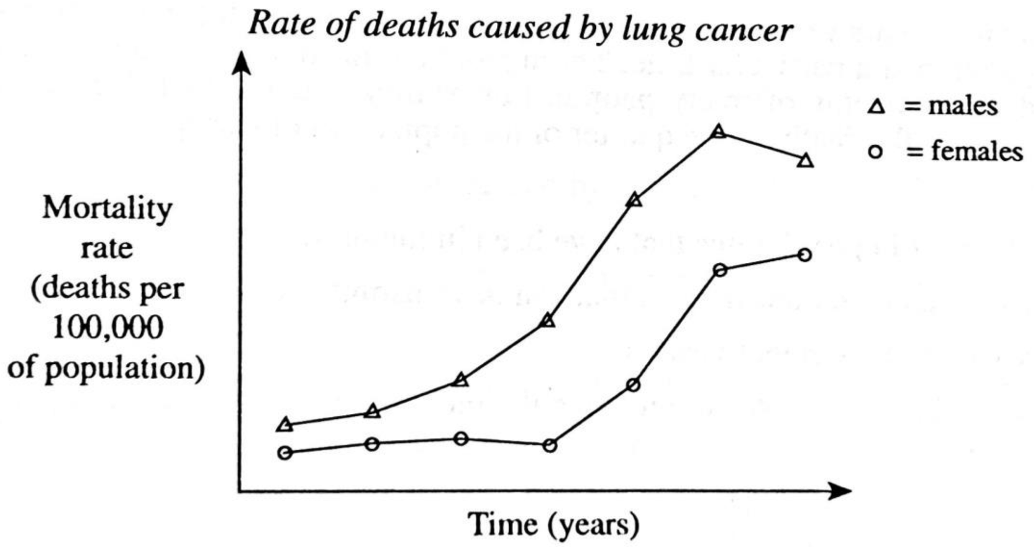
(A) Sunken stomata

(B) Large leaves for cooling

(C) Stomata on the upper surface of the leaves

(D) Large hard leaves with a high density of stomata

14 The following graph illustrates the changes in the rate of deaths caused by lung cancer.



What would be the best conclusion to make from this graph?

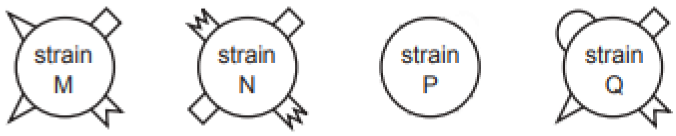
(A) A greater percentage of women die from lung cancer than men.

(B) In the past, women smoked less tobacco and lived longer than men.

(C) The number of deaths due to lung cancer is greater for men than women.

(D) The number of men dying from lung cancer is decreasing because fewer men are smoking.

15 The following diagram shows the different forms of antigens, and their relative distribution, found on the surfaces of different strains of *Staphylococcus* bacteria.



You wish to make a single vaccine that will be effective against as many strains as possible of the Staphylococcus bacteria shown. Which strain would you use to make the vaccine?

(A) Strain M

(B) Strain N

(C) Strain P

(D) Strain Q

16 Which of the following is not considered a biological mutagen?

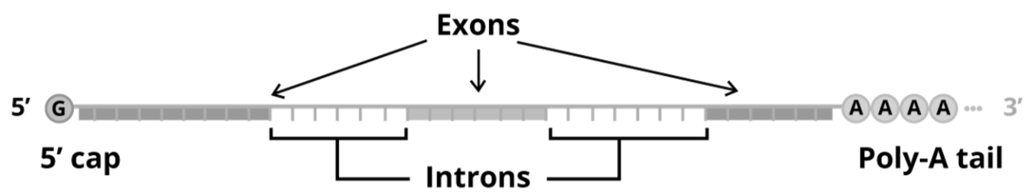
(A) Viruses

(B) UV radiation

(C) Preservatives such as nitrites

(D) Mycotoxins found in mouldy food

17 An intron is a long stretch of non-coding DNA found between exons (or coding regions) in a gene. Genes that contain introns are known as discontinuous or split genes as the coding regions are not continuous. Introns are found only in eukaryotic organisms.



Which of the following statements is correct?

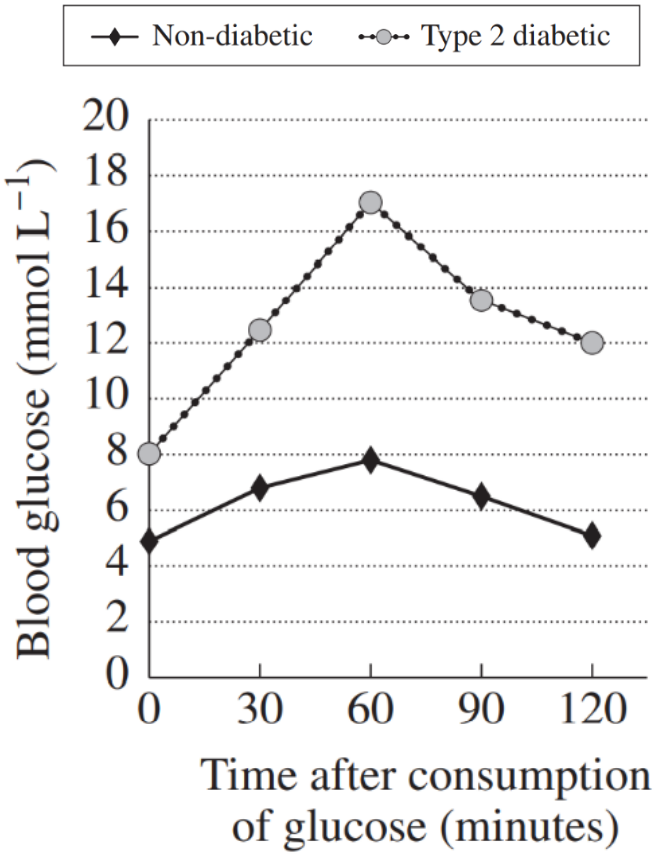
(A) The non-coding regions of DNA have no function.

(B) Introns contain the code for synthesising proteins.

(C) The resulting mRNA is shorter than the gene in eukaryotes.

(D) Processing the mRNA into the final molecule used in translation causes mutations.

18 The graph below compares the blood glucose concentration in a non-diabetic person with a person who has diabetes.



Based on this information, what is the most probable cause of diabetes?

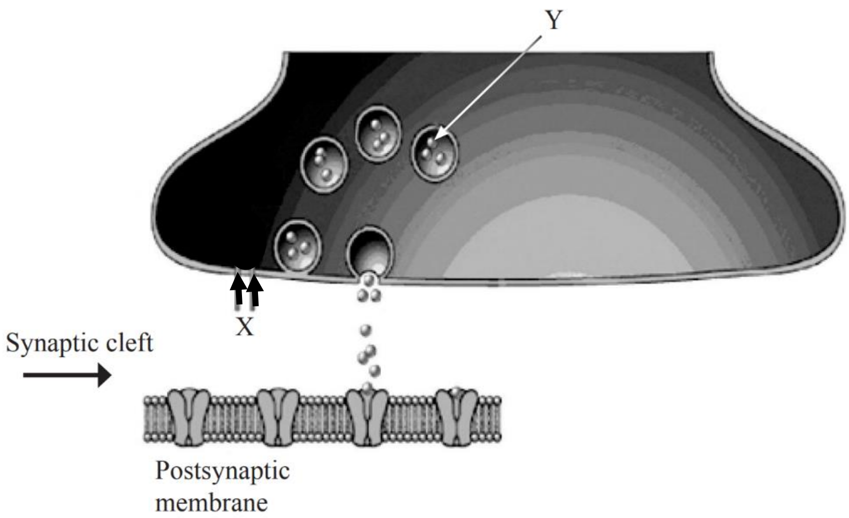
(A) Increased responsiveness of the body to insulin.

(B) Decreased responsiveness of the body to insulin.

(C) Beta cells in the pancreas producing too much insulin.

(D) Alpha cells in the pancreas producing not enough glucagon.

19 In the diagram of synaptic transmission below, what is indicated by the letters X and Y?



	X	Y
(A)	Direction of synapse	ions
(B)	Axon terminal	Neurotransmitter
(C)	Direction of synapse	Neurotransmitter
(D)	Dendritic terminal	ions

20 In an isolated small population of mice there are two alleles present for a particular gene. The frequency of the functional allele is 0.8, while the frequency of the nonsense mutant allele is 0.2. One generation later, the allele frequencies are 0.6 for the functional allele and 0.4 for the nonsense allele.

Which evolutionary process explains the change in allele frequency?

(A) Mutation

(B) Genetic drift

(C) Genetic flow

(D) Natural selection

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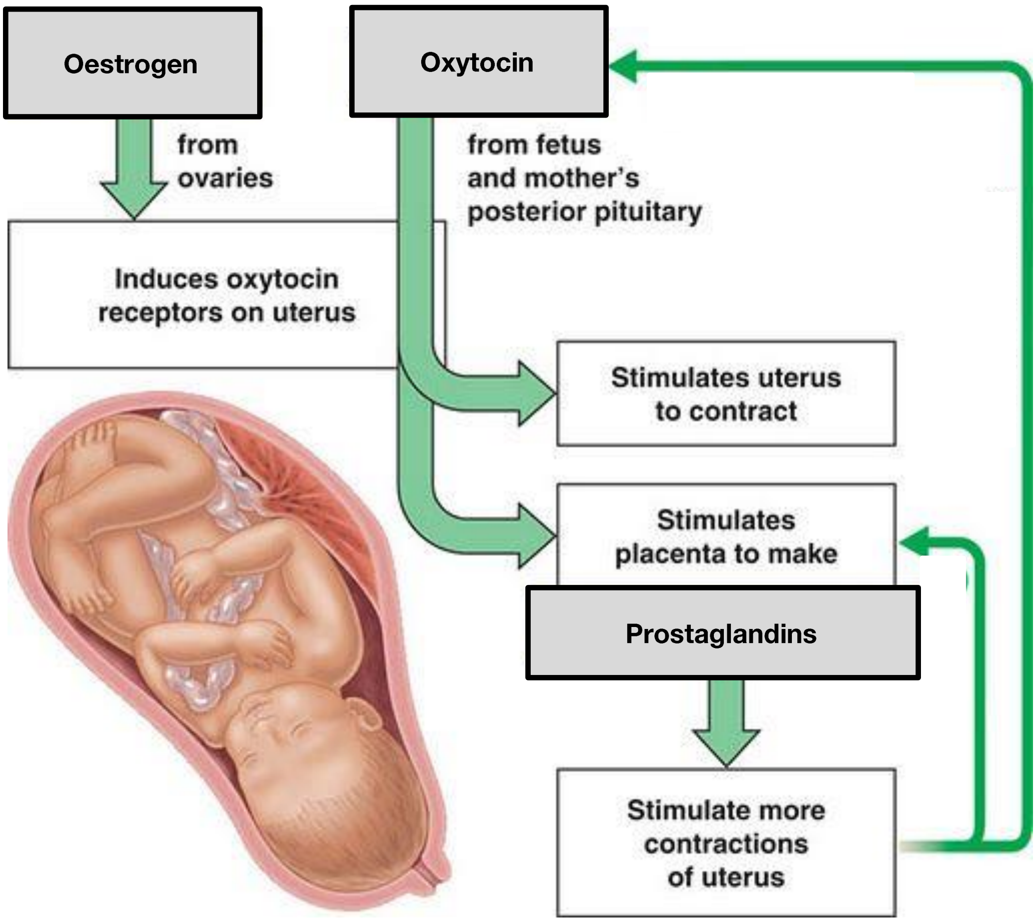
SECTION II: SHORT AND LONG ANSWER

Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER

Question 21 (5 marks) **Marks**

Look at the diagram below which shows the hormone interactions that stimulate labour leading to the birth of a baby.



(a) Explain how this feedback loop causes an increase in contractions during the birthing process. **2**

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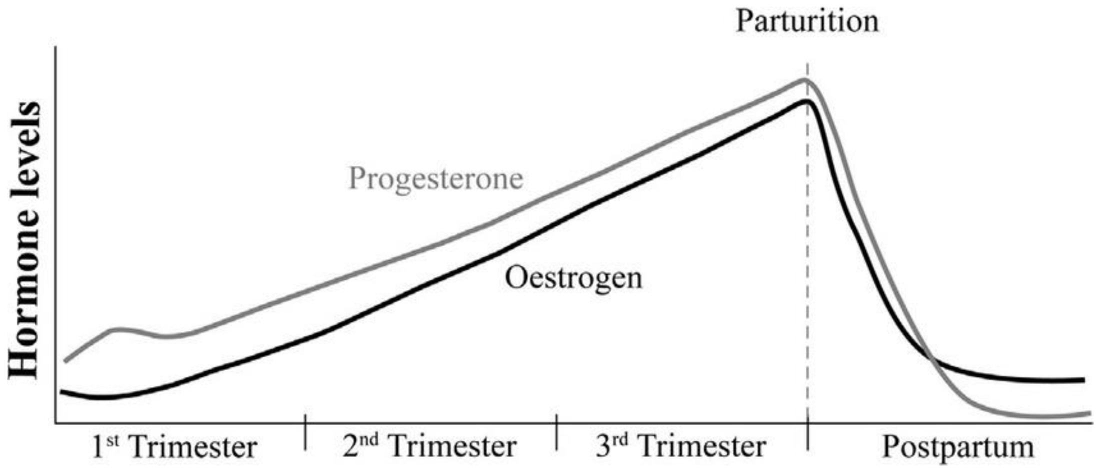
Question 21 continued on next page.

Question 21 continued. **Marks**

(b) Analyse the graph below which shows the change in oestrogen and progesterone hormone levels during the three trimesters of pregnancy and following birth (parturition).



Using the information above and your knowledge, explain why progesterone supplements are given to women with a history of early labour and premature births.

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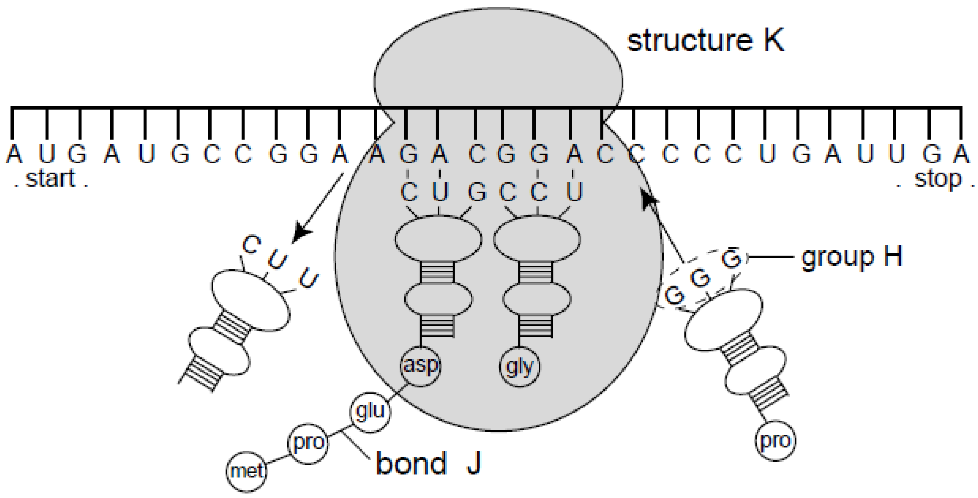
Question 22 (6 marks) **Marks**

The diagram below shows a process that is part of protein synthesis.



(a) Identify the process shown in this diagram.

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(b) Explain the importance of the structure labelled 'group H' in this process.

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(c) Discuss how a specific type of point mutation would affect the outcome of the process in the diagram above.

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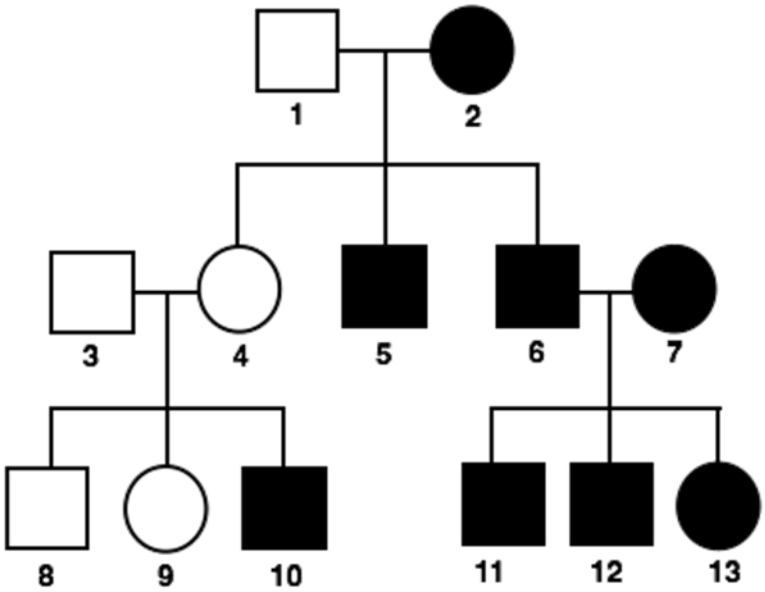
Question 23 (5 marks) **Marks**

Look at the pedigree below showing an inherited, autosomal condition in a family.



(a) Is this condition dominant or recessive? Justify your answer with evidence from the pedigree.

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(b) Individual 13 marries a man who does not have the condition.

i. Identify the possible genotype(s) of the husband of individual 13.

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Question 23 continued on next page.

Question 23 continued. **Marks**

ii. Determine the probability of this couple having a child **with** the condition.

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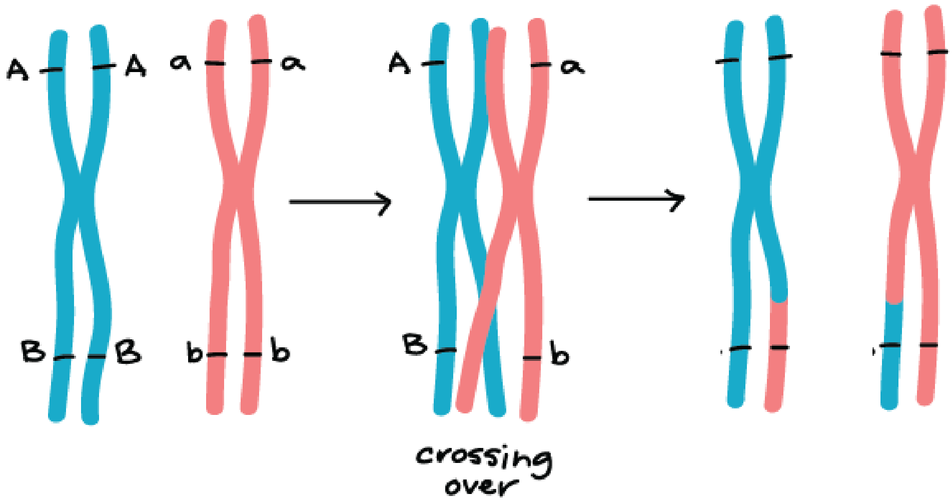
Question 24 (4 marks)

Below is an incomplete diagram, modelling the process of crossing over. Two genes are shown (A and B). A represents hair colour. B represents eye colour.



(a) Complete the diagram by adding the appropriate alleles to the last pair of chromosomes.

1

(b) Identify the **genotype(s)** of an individual who would show the recessive phenotype for hair colour and the dominant phenotype for eye colour.

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Question 24 continued on next page.

Question 24 continued. **Marks**

(c) If eye colour was a co-dominant condition, how would the phenotype for a heterozygous individual differ from the phenotype in part (b)?

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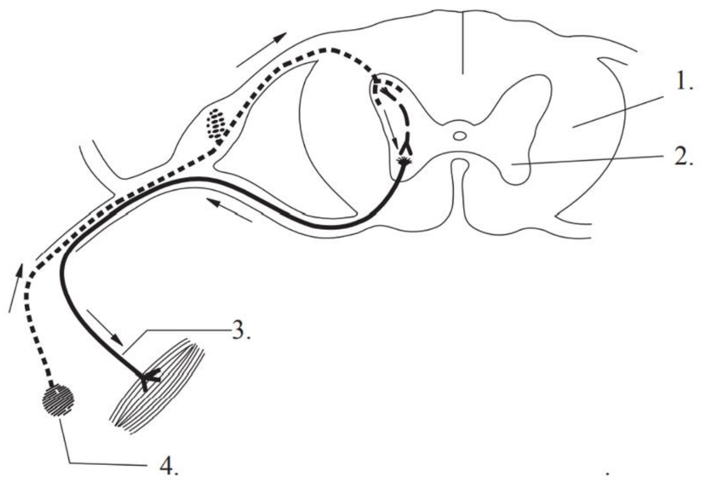
Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

CANDIDATE NUMBER							

Question 25 (8 marks) **Marks**

The diagram below represents the spinal cord in cross section.



(a) Using a number from the diagram, identify:

i. The receptor……………………………………………………………………

ii. The motor (effector) neuron…………………………………………………. **2**

Question 25 continued on next page.

Question 25 continued. **Marks**

(b) Describe how the message is sent from the central nervous system to the effector. **3**

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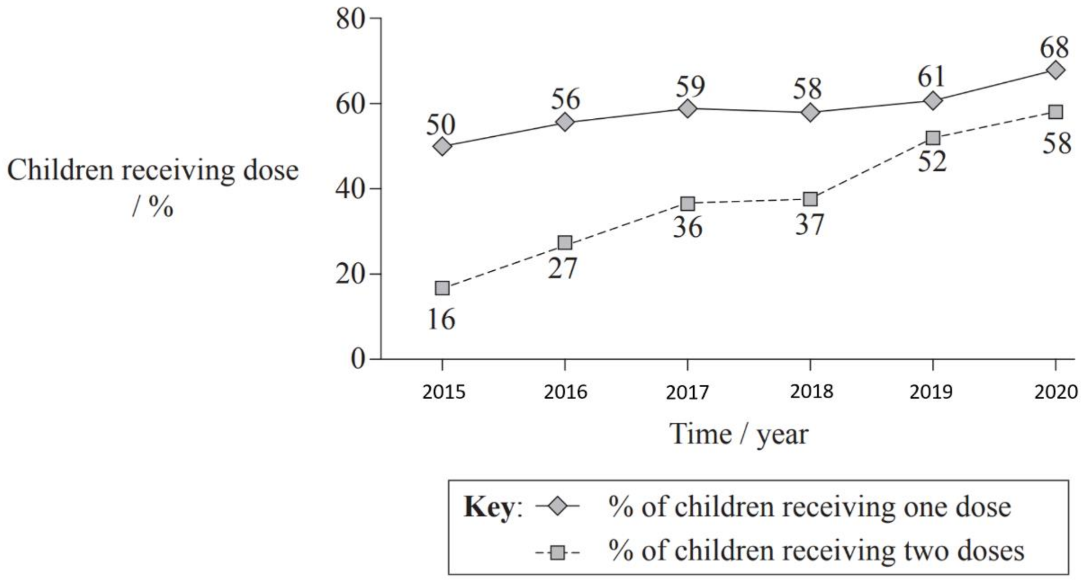
(c) In the table below, compare the nervous system with the endocrine system. **3**

Nervous System	**Endocrine System**

Question 26 (12 marks) **Marks**

Retinol (vitamin A) is essential for healthy development in children. It is an integral component in the photosensitive cells in the retina of the eye. The United Nations Children's Emergency Fund (UNICEF) estimates that at least 70% of children around the world need retinol supplements to prevent any developmental problems.

The graph below shows the percentage of children (indicated above each point) receiving one or two doses of retinol per year between 2015 and 2020.



(a) For which 2 consecutive years is the greatest increase shown in the number of children receiving the two-dose supplement?

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(b) Identify another factor that should be considered when analysing this data.

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(c) Analyse the trend in the proportion of children receiving retinol from 2015 to 2020.

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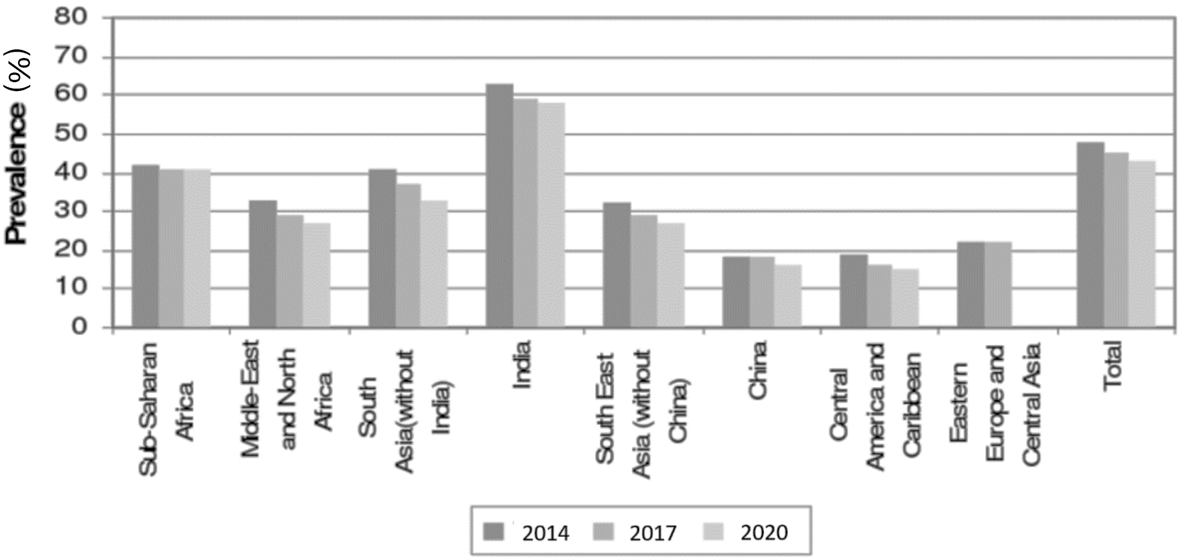
Question 26 continued on next page.

Question 26 continued. **Marks**

The World Health Organization (WHO) has classified vitamin A deficiency as a public health problem. In 2014 it affected almost half of the children worldwide.

The graph below shows the prevalence of vitamin A deficiency in children (6 months to 19 years old) in different regions of the world.



(d) Comment on the trend in worldwide prevalence from 2014-2020 and account for the difference observed.

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(e) Calculate the population of children with vitamin A deficiency in India in 2017, if the total children population was 253 million.

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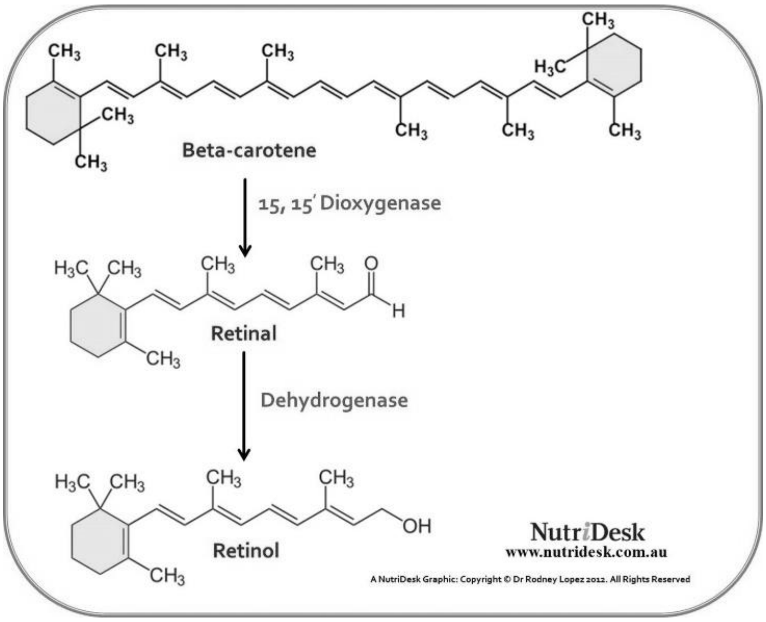
Question 26 continued on next page.

Golden Rice is a variety of rice that has been genetically modified. The genes for the synthesis of Beta-carotene have been taken from the daffodil and inserted into the genome of a strain of rice.

Beta-carotene is an organic, pigment found in plants and fruits. It is converted in the liver in the biochemical pathway shown below:



(f) The implementation of Golden Rice into the rice fields of rural India is slowly becoming a reality in an attempt to combat vitamin A deficiency in the country.

Using all the information provided in Question 26, discuss the potential of Golden Rice reducing vitamin A deficiency in India. **4**

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Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

Question 27 (2 marks) **Marks**

Compare a chemical response of animals and plants to prevent infection.

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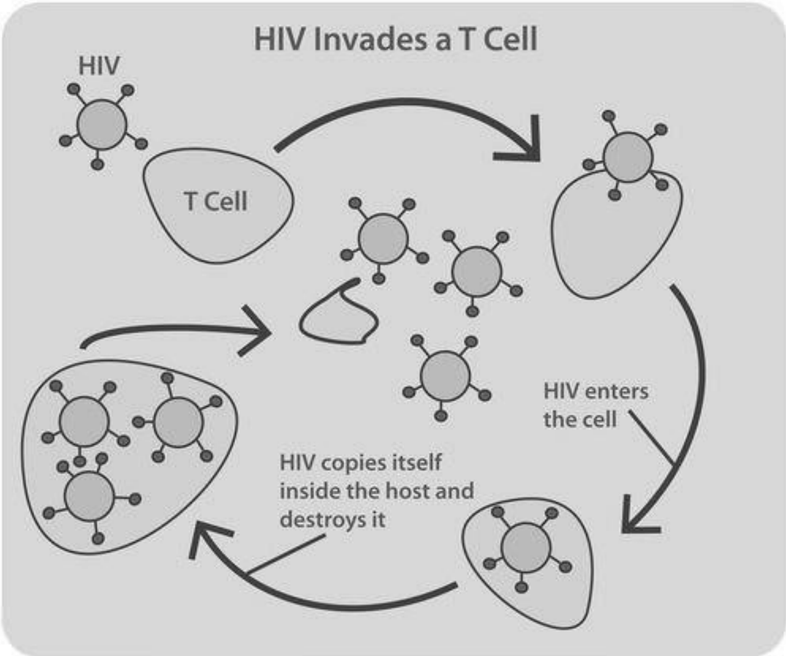
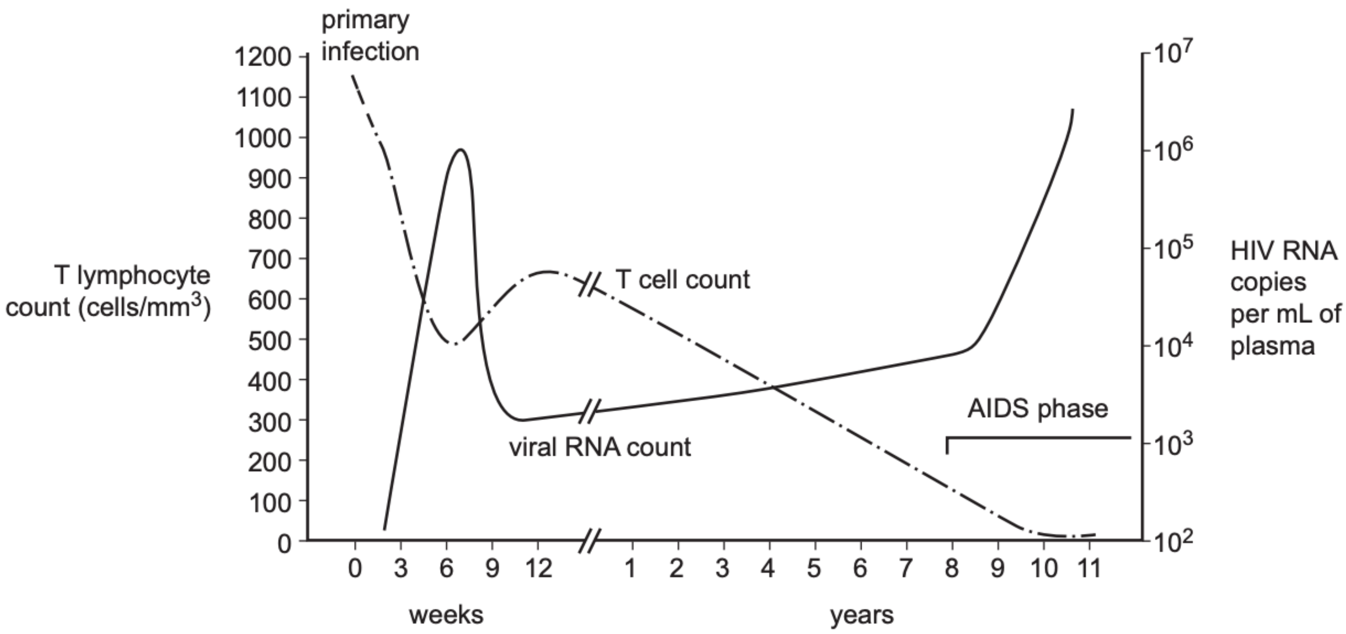
Question 28 (11 marks) **Marks**

Look at the stimuli below which depicts how HIV (human immunodeficiency virus) invades and reproduces in the body of a patient.



HIV RNA count in the blood and the effect of the virus on the number of specific immune cells



Source: adapted from Paola Paci et al., 'A Discrete/Continuous Model of Anti-HIV Response and Therapy', conference paper, Tenth International Conference on Computer Modeling and Simulation, May 2008, p. 484

Question 28 continued on next page.

Question 28 continued. **Marks**

(a) The HIV pathogen does not have any ribosomes. How does it make new viral proteins to replicate itself?

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(b) From 0 to 6 weeks, how much has the T lymphocyte count (cells/mm^3) reduced in the patient?

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(c) Using the information provided above and your own knowledge, explain why HIV patients who develop AIDS (Acquired immunodeficiency syndrome) can die from a common cold.

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Question 28 continued on next page.

Question 28 continued. **Marks**

(d) HIV is spread through body fluids including blood, semen, vaginal fluids, anal mucus and breastmilk, and from mother to child. Describe measures to prevent the spread of this disease and justify your choices. **5**

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Question 29 (8 marks) **Marks**

"Genetic technologies have the potential to eliminate infectious disease."

Evaluate this statement using examples. In your answer you should explain why techniques used to prevent disease have changed and assess the influence of economic, social and cultural factors.

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Answer the questions in the spaces provided.
Show all relevant working in questions involving calculations.

Question 30 (19 marks) **Marks**

Tasmanian Devils were once abundant on mainland Australia as evidenced by fossil remains. It is thought that they became extinct on the mainland about 400 years ago and are now only found in the forests of Tasmania. It is at risk from the contagious devil facial tumour disease (DFTD, see Figure 1), which originated by mutation in a female devil about 25 years ago. Devil numbers have been reduced to less than 10 percent of the original population (Figure 2). It has been hypothesised that one of the reasons why this disease has spread so readily is the low genetic diversity of the population.

Fig. 1 Tasmanian devil with facial tumour

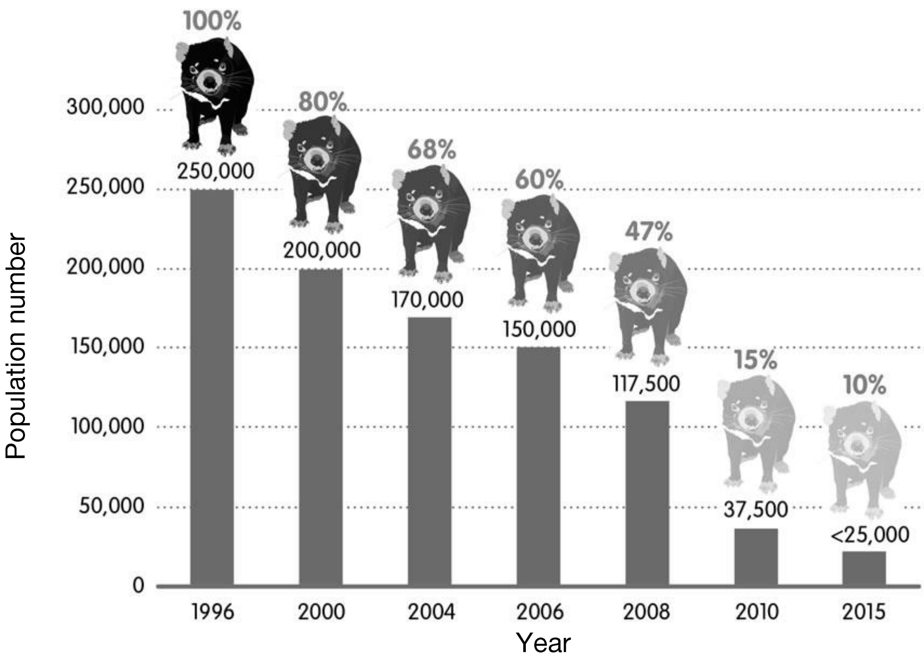

Fig. 2 Change in Tasmanian devil population

Question 30 continued on next page.

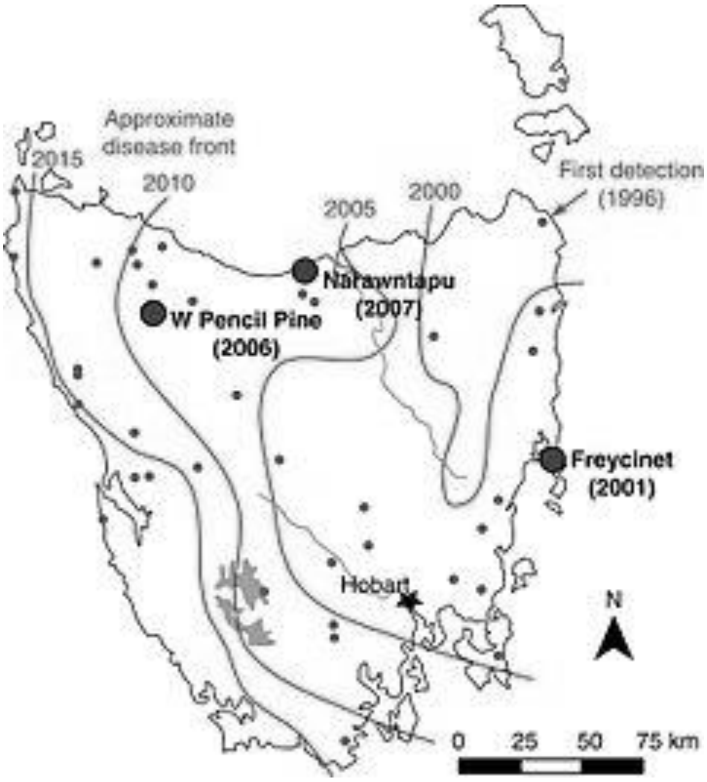

Fig. 3 Spread of DFTD over time.

The facial cancer spread rapidly across the island (Figure 3) because nearly all devils have very similar Class I major histocompatibility complexes (MHC I) – unique molecules on the surface of a cell that identify the cell as belonging to "self", allowing the immune system to recognise foreign molecules. When devils fight, cancer cells get directly transferred from an infected devil to wounds in an uninfected devil as a result of biting injuries. Because of their similar MHCs, the immune system doesn't recognise the transferred cells as being different and the cancer easily spreads.

Recent evidence suggests that the devil population may have developed some genetic resistance to the disease in just four to six generations. It has been hypothesised that there are alleles present in some devil populations that may be helping the immune system to recognise the cancer cells as foreign cells.

To help ensure the long-term survival of this species, a captive breeding program was established in 2011 on the mainland at Barrington Tops, NSW. From the original population of 44 DFTD-free devils, over 300 offspring have been born. The aim is to eventually release some of these animals back into the wild in Tasmania.

Question 30 continued on next page.

 Marks

(a) Instances of transmissible cancers are extremely rare. In humans, once a cancer cell is recognised, describe the response of the immune system.

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(b) Evaluate whether DFTD should be classified as either an infectious or non-infectious disease.

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Question 30 continued on next page.

Question 30 continued. **Marks**

(c) Outline how biotechnology can be used to generate useful population genetics data for the Tasmanian devil, and with reference to the information detailed above in Question 30, explain how this data can be used to assist in the conservation management of this species. **8**

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Question 30 continued on next page.

Question 30 continued.

Marks

Question 30 continued on next page.

Question 30 continued. **Marks**

Question 30 continued on next page.

Question 30 continued. **Marks**

(d) Evaluate the likely effects of mutation, gene flow and genetic drift on the gene pool of the Tasmanian devil over the last 400 years. **5**

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END OF PAPER

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Section II extra writing space

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